Inelastic Capital in Intangible Economies*

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Abstract

Capital in modern economies increasingly takes the form of intangible capital, whose formation heavily depends on the contributions of specialized workers—such as inventors, managers, and entrepreneurs. To examine the macroeconomic implications of this fact, we develop and calibrate a general neoclassical model where capital formation requires both investment goods (tangible investments) and specialized labor (intangible investments). We show that rising intangibles renders the supply of capital more inelastic owing to the limited supply of specialized labor. Rising intangibles also change the incidence of capital taxation: whereas in traditional neoclassical models the tax burden falls entirely on production workers, in intangible economies, it is borne primarily by specialized workers and capital owners.

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Introduction

The standard neoclassical growth model treats capital formation as a frictionless process: final goods can be converted into new capital at no cost. This assumption implies that this model predicts an infinitely elastic supply curve of capital. As a result, shifts in the supply or demand for capital — arising from changes in household saving or changes in the productivity of investment — result in large movements in quantities rather than prices.

This prediction, however, stands in stark contrast to the empirical data, as asset prices exhibit larger volatility than the capital stock. While models with capital adjustment costs yield a short-run inelastic capital supply (Hayashi, 1982), they still imply an infinitely elastic capital supply in the long run. Thus, these models cannot explain why "savings gluts" (Bernanke et al., 2007; Mian et al., 2020) tend to generate higher movements in the price than in the quantity of assets.[1]

This paper takes a different route: we emphasize that transforming final goods into productive capital requires specialized labor—entrepreneurs, inventors, and managers—whose aggregate supply is inherently limited. Because more capital cannot simply be constructed from final goods but must also be conceptualized, designed, and implemented by these scarce workers, the supply curve for capital slopes upward over any horizon.

This more realistic view of capital formation also highlights a key link between financial and labor markets. In our model, an influx of savings raises the wages of specialized investment workers relative to those of production workers. This mechanism suggests that the secular decline in interest rates has played an important role in driving the rise in wage polarization. (Autor and Dorn, 2013). Finally, this perspective on capital formation has crucial implications for the incidence of corporate taxation: while the tax burden falls entirely on production workers in the standard neoclassical model, it is borne primarily by specialized workers and capital owners in intangible economies.

Our paper proceeds as follows. We first develop a stylized extension of the standard neoclassical growth model that features a more realistic investment function. Our key departure is that the investment good is more labor intensive than the consumption good, reflecting the fact that modern capital formation requires specialized labor (i.e., inventors, entrepreneurs, financiers).

We then characterize the capital supply elasticity in the model; that is, the responsiveness of the quantity of capital to changes in the price of capital.[2] In the neoclassical growth model, capital supply is infinitely elastic, as the final good can be seamlessly transformed into capital. While adjustment costs (Hayashi, 1982) may reduce this elasticity in the short run, they do not alter the long-run elasticity, which remains infinite. In contrast, we show that, if the investment sector is more labor-intensive than the consumption sector, the capital supply elasticity remains finite, even in the long run.

[1] See, for instance, Gutiérrez and Philippon, 2017 and Crouzet and Eberly, 2023 for empirical evidence.

[2] To be precise, the supply elasticity of capital in the model is the elasticity of the capital stock (at some horizon) with respect to a change in the value of capital (i.e., the marginal q).

In Section 2, we use data from the national accounts to study the factor intensity of the consumption and investment goods sectors. We combine sectoral production data from KLEMS with the BEA's input-output (I-O) make-use matrix to compute the consolidated labor share of each sector. This approach accounts for the labor embedded both directly and indirectly across industries. Our findings reveal a growing wedge in labor intensity between the investment and consumption sectors — particularly in the use of high-skilled labor. As a robustness check, in the appendix, we show that one would obtain similar results using firm-level accounting data from Compustat.

In Section 3, we develop a more quantitative model to quantify the supply elasticity of capital in the data. We add four ingredients to the stylized model: (i) multiple types of labor and capital, (ii) type-specific labor supply elasticity, (iii) type-specific internal adjustment costs, and (iv) a tax on capital payouts. The model, disciplined on the micro-data, implies a long-run supply elasticity of capital of 1.5: that is, a permanent shift in the demand for capital (i.e. decline in interest rates) that rise q by 1% would lead to a 1.5% increase in the long-run quantity of capital.

Finally, we use the calibrated model to quantify the incidence of corporate taxes in our economy. In the neoclassical growth model, the capital supply elasticity is infinite. As a result, capital tax cuts have a large effect on investment, and end up being born by production workers via higher wages. In intangible economies, however, tax cut have a weaker effect on investment, hence a lower effect on production worker wages. As a result, capitalists and investment workers benefit more from lower taxes via higher returns (for capitalists) and higher wages (for investment labor). In the calibrated model, the incidence of capital taxes is roughly two-thirds for capitalists, a quarter for workers in the investment good sector, and one-twelfth for workers in the consumption good sector.

Literature review. At the core, our theory builds on the neoclassical two-sector model (see, e.g., Uzawa, 1961; Srinivasan, 1964), in which the consumption and investment goods have different production functions. This leads to the presence of external adjustment costs, i.e., the situation where equilibirum changes in prices (in our case wages) dampen the equilibrium response of investment to demand shocks. As discussed in Mussa (1977), this channel differs from external adjustment costs (see, e.g., Lucas Jr, 1967; Uzawa, 1969). Our main contribution is to quantify the importance of these external adjustment costs, in the context of a modern economy that features labor-intensive intangible investments.

A growing literature documents a rise in intangible capital (see, e.g., Eisfeldt and Papanikolaou, 2013; Peters and Taylor, 2017; Eisfeldt, Falato and Xiaolan, 2023; Corrado, Haskel, Jona-Lasinio and Iommi, 2022). Beyond being harder to measure, intangible capital may exhibit distinct economic properties relative to tangible capital. In Eisfeldt and Papanikolaou (2013), intangible capital is unique because it is embedded in key talents. In Crouzet, Eberly, Eisfeldt and Papanikolaou (2022), intangible capital tends to be non-rival—allowing it to be used simultaneously in different production streams—and is characterized by limited excludability, which prevents firms

from capturing all the associated benefits or rents. In this paper, we shift our focus away from the economic properties of intangible capital to its distinct inputs, namely, that it is created by specialized labor in finite supply.

The core idea that specialized labor is a key input in capital formation broadly (including innovation, business formation, etc) appears is several leading models (i.e., Romer, Hopenhayn, Melitz, etc) or more recently in models than emphasize managerial and entrepreneurial inputs (see, e.g., Luttmer, 2018 and Kogan et al., 2020).

Finally, our paper offers an alternative explanation for investment stagnation despite high Tobin's Q inferred from the data. The existing literature focuses on markups and market power (e.g., Gutiérrez and Philippon, 2017; Crouzet and Eberly, 2019; Barkai, 2020; Ball and Mankiw, 2023; De Ridder, 2024). In contrast, we emphasize the fact that capital formation requires specialized labor (i.e., entrepreneurs, inventors, etc.) which constrains the extent to which investment goods can be transformed into productive capital. Put differently, in our paper, the "fixed factor" that limits the ability (or willingness) of firms to scale is not the fact that it would lower their price, but, rather, that it would increase their labor cost.

1 Stylized model

We now characterize the long-run (deterministic) equilibrium in a two-sector model with heterogeneous factor shares, which follows closely Uzawa (1961).

Technology. There are two factors of production (capital K and labor L) which are used in two final-good producing sectors (consumption C and investment I). The technological environment is characterized by two production functions and a capital accumulation equation:

$$C = K_C^{\alpha} L_C^{1-\alpha}, \qquad \text{(consumption good sector)}$$
$$I = K_I^{\beta} L_I^{1-\beta}, \qquad \text{(investment good sector)}$$
$$K' = (1-\delta)K + I. \qquad \text{(capital accumulation)}$$

Here, in the rest of the paper, we use the apostrophe symbol to represent the value of a quantity in the next period. In both sectors, the final good is produced using a Cobb-Douglas production function, with capital share $\alpha \in (0,1)$ in the consumption good sector and $\beta \in (0,1)$ in the investment good sector. In the special case $\alpha = \beta$, we recover the neoclassical growth model.

Optimality conditions. We consider a small open economy with interest rate r. We assume that each sector is a price taker in the input market (the wage w and rental rate of capital r_K are taken as given) as well as in the output market (the relative price of capital q is taken as given). Firm

optimization implies the following optimality conditions:

$$w = (1-\alpha)(C/L_C), \qquad \text{(labor demand in C)}$$
$$w = (1-\beta)(qI/L_I), \qquad \text{(labor demand in I)}$$
$$r_K = \alpha(C/K_C), \qquad \text{(capital services demand in C)}$$
$$r_K = \beta(qI/K_I). \qquad \text{(capital services demand in I)}$$

Note that the marginal product of each factor (labor or capital) is equalized across sectors.

The demand side of the model is characterized by an inelastic supply of labor $\overline{L}$ and an arbitrage equation

$$r = (r_K' + q')/q - \delta. \qquad \text{(arbitrage equation)}$$

The arbitrage equation originates from the demand-side in the capital market (i.e., from the perspective of a household buying capital), and implies that holding capital must yield a net return r.

A long-run equilibrium is an allocation $(K_C, K_I, K, L_C, L_I, L)$ and prices (w, r_K, q) such that firms and households optimize, the steady-state condition $\delta K = I$ holds, and markets clear:

$$K_C + K_I = K, \qquad \text{(capital services market clearing)}$$
$$L_C + L_I = L. \qquad \text{(labor market clearing)}$$

Capital market equilibrium. The capital market equilibrium in the model, which pins down the quantity K and price q of capital in the long-run, can be expressed as a standard supply and demand diagram. To be precise, the "capital supply curve" is the locus of values (q, K) that satisfy all long-run equilibrium conditions except (arbitrage equation). Mathematically, we drop one endogenous variable (q) and remove one equation. Economically, the capital supply curve governs the long-run level of capital formation *given a relative price of capital* q.

The "capital demand curve" is the locus of values (q, K) that satisfy the arbitrage equation. The demand curve states that households want to pay a lower price q for a unit of capital when the return on capital r_K is low. It is merely a valuation equation. The following proposition characterizes the first-order dynamics around the capital market equilibrium (i.e., we compute the supply and demand elasticities around the steady-state).

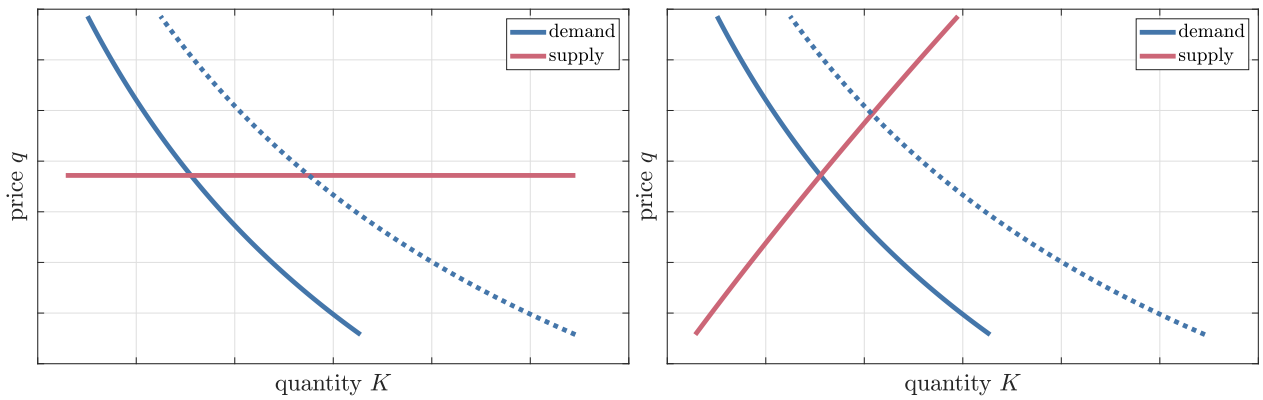
Proposition 1.1 (Elasticities)**.** *Movements along the capital supply/demand curves are characterized by*

$$\frac{\partial \log K^s}{\partial \log q} = \frac{1}{\alpha-\beta} - \frac{(1-\beta)\delta}{(1-\alpha)r + (1-\beta)\delta}, \qquad \text{(Supply elasticity)}$$
$$\frac{\partial \log K^d}{\partial \log q} = -\left(\frac{1}{1-\alpha} + \frac{(1-\beta)\delta}{(1-\alpha)r + (1-\beta)\delta}\right), \qquad \text{(Demand elasticity)}$$

around the steady-state (q, K).

Proof. See Appendix A. □

The capital supply elasticity depends crucially on the relative labor share in the investment sector $\alpha - \beta$.[3] The idea is that, when the relative labor share is high (i.e., investment goods are relatively labor intensive), then a rise in investment puts upward pressure on equilibrium wages, which disproportionately affects the investment sector. This phenomenon is usually referred to as "external adjustment costs", due to the fact that equilibrium price responses dampen the response of investment to a change in q. When both sectors have the same labor share $\alpha = \beta$, as in the neoclassical growth model, the capital supply elasticity is infinite, reflecting the fact that the investment and consumption goods must have the same price.



(a) Neoclassical growth model ($\alpha = \beta$).

(b) General case $\alpha > \beta$.

Figure 1: Long-run capital market equilibrium

Ultimately, the slope of the capital supply curve matters because it affects how demand shocks in the capital market transmit to prices and quantities. Figure 1 characterizes the effect of an arbitrary demand shock on the long-run equilibrium via a diagram. In the special case $\alpha = \beta$, the supply curve is horizontal (i.e., the price of capital goods relative to consumption goods is constant). In that case, demand shocks transmit entirely to quantities, not prices. In contrast, when investment is relative $\alpha > \beta$, the capital supply elasticity is finite, which means that demand shocks are partly absorbed by changes in q.

Case study: capital income tax cut. Consider a simple extension of the model where there is a tax rate τ on capital income $r_K K$. This represents a pure demand shock: it only enters the demand-side of the model via (arbitrage equation).

What is the long-run effect of a permanent capital income tax cut around the undistorted steady-state (i.e., $\tau = 0$)? Using standard equilibrium calculations using supply and demand

[3] Since the labor share in the consumption and investment sectors are respectively $1 - \alpha$ and $1 - \beta$, the relative labor share is therefore $1 - \beta - (1 - \alpha) = \alpha - \beta$.

elasticities from Proposition 1.1, we can express the first-order as:

$$\mathrm{d}\log K = \frac{\partial \log K^s}{\partial \log q} \cdot \mathrm{d}\log q \tag{supply}$$

$$\mathrm{d}\log K = \frac{\partial \log K^d}{\partial \log q} \cdot \mathrm{d}\log q + \frac{1}{1-\alpha}\,\mathrm{d}\log(1-\tau) \tag{demand}$$

The equilibrium response to the shock is:

$$\frac{\mathrm{d}\log q}{\mathrm{d}\log(1-\tau)} = \frac{\alpha-\beta}{1-\beta}, \tag{price effect}$$

$$\frac{\mathrm{d}\log K}{\mathrm{d}\log(1-\tau)} = \frac{1}{1-\beta} - \frac{(\alpha-\beta)\delta}{(1-\alpha)r+(1-\beta)\delta}. \tag{quantity effect}$$

The equilibrium response of prices q depends crucially on the excess labor share in the investment sector $\alpha-\beta$. In the NGM, the price response is zero, while in the empirically relevant case $\alpha > \beta$, the response is positive. The reason is that the marginal cost of producing capital goods increases more than consumption goods, given the fact that investment goods are more labor intensive. The response of quantities K is the sum of two terms. First, the term $-\frac{1}{1-\alpha}$ is exactly as in the NGM, and reflects the fact that the marginal product of capital declines with K. The second term is negative in the empirically relevant case: it dampens the response of capital accumulation due to external adjustment costs. See Appendix A for the derivations.

Denoting wealth as the value of the capital stock $W = qK$, we can sum the price and quantity effects to obtain

$$\frac{\mathrm{d}\log W}{\mathrm{d}\log(1-\tau)} = \frac{1}{1-\alpha} - \frac{\alpha-\beta}{1-\beta}\left(\frac{\alpha}{1-\alpha} + \frac{(1-\beta)\delta}{(1-\alpha)r+(1-\beta)\delta}\right)$$

Notice that, in the empirically relevant case $\alpha > \beta$, the response of wealth to the tax cut is lower than the response in the neoclassical growth model (i.e., $\frac{1}{1-\alpha}$). However, as we just established, a larger share of the rise in wealth is due to change in valuations $\mathrm{d}q$ rather than changes in capital formation $\mathrm{d}K$. These types of revaluation effects have important redistributive effects that affect both inequality and welfare (see, e.g., Gomez and Gouin-Bonenfant, 2024; Fagereng, Gomez, Gouin-Bonenfant, Holm, Moll and Natvik, 2024).

2 Empirical evidence

The central idea of the stylized model is that producing capital goods is more labor intensive—particularly in terms of specialized labor—than producing consumption goods. We now provide empirical evidence that this pattern increasingly holds in the U.S. economy, driven in part by the rising importance of intangible investment.

2.1 Datasets

We begin with the BEA's input-output matrix, available from 1963 to 2020, which provides a detailed decomposition of expenditures across 71 industries (U.S. Bureau of Economic Analysis, 2024a). This matrix tracks how each industry allocates its spending across intermediate inputs, labor, and capital. We further disaggregate the labor component into payments to college-educated and non-college-educated workers using data from the Integrated Industry-Level Production Account (U.S. Bureau of Economic Analysis, 2024b). In addition, the input-output matrix reports how each industry's output is allocated—whether to other industries as intermediate goods, or to final uses in consumption and investment.

In a series of revisions to the National Income and Product Accounts in 1999, 2013, and 2018, the U.S. Bureau of Economic Analysis reclassified firm expenditures on intellectual property products—software, research and development, and artistic originals—as investments. Corrado et al., 2009 argues for a much broader definition of intangible capital. Following their lead, we reclassify certain firm expenditures on professional services as investments.

Specifically, we capitalize 60% of firm spending on the following professional service categories: "Administrative and support services", "Miscellaneous professional, scientific, and technical services", and "Management of companies and enterprises".[4] The first category reflects spending on outsourced business support services, such as personnel administration and training. The second includes externally purchased services like accounting, consulting, design, and computer services. The third captures the service output of establishments that manage other units within a company. Purchases of these services plausibly represent investment in what the literature refers to as organization capital (Atkeson and Kehoe, 2005; Eisfeldt and Papanikolaou, 2013; Crouzet and Eberly, 2021; Crouzet et al., 2022)—expenditures related to workforce human capital, distribution systems, logistics, product design, and customer and brand capital. We group all of these under the umbrella term "Organizational services."

Figure 2 shows the composition of investment as a share of GDP after doing this reclassification. We observe a decline in investment in structures and equipment, and a corresponding rise in investment in intellectual property products and organizational services.

2.2 Methodology

Our goal is to compute the labor share of value added required to produce one unit of a consumption good versus one unit of an investment good.

We proceed in two steps. First, we compute the consolidated labor share of each industry. This measure accounts not only for the share of the industry's own value added paid to labor, but also for the labor embodied in all intermediate inputs used in its production. Formally, we use the

[4] Ideally, we would use a more detailed industry breakdown, but the BEA input-output matrix is available only at the level of 71 industries for a time series beginning in 1963.

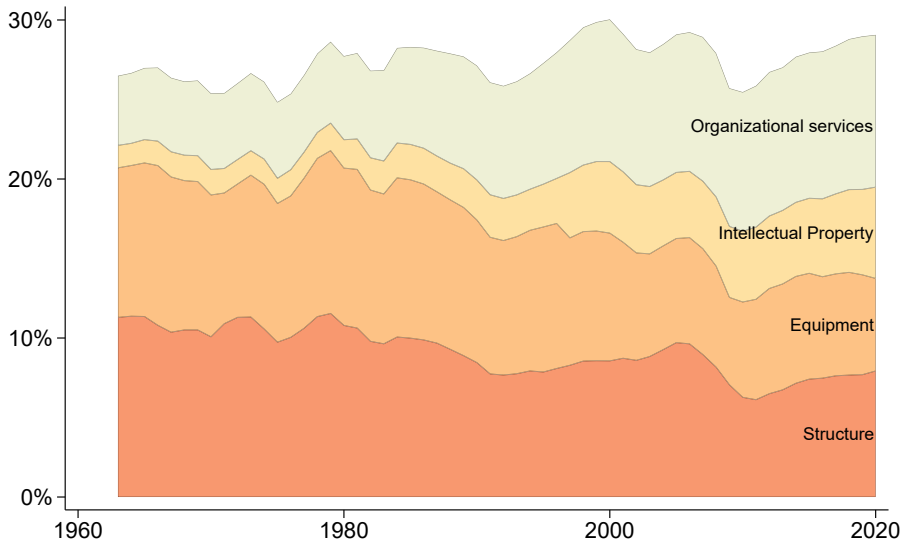


Figure 2: Types of investment expenditures as a fraction of GDP

Leontief inverse of the input-output matrix to capture both direct and indirect contributions of labor throughout the production network.

Second, we compute the labor share of the consumption sector as the weighted average of industry labor shares, where the weights correspond to the share of final consumption expenditures coming from each industry. Similarly, the labor share of the investment sector is calculated using the share of final investment expenditures as weights.

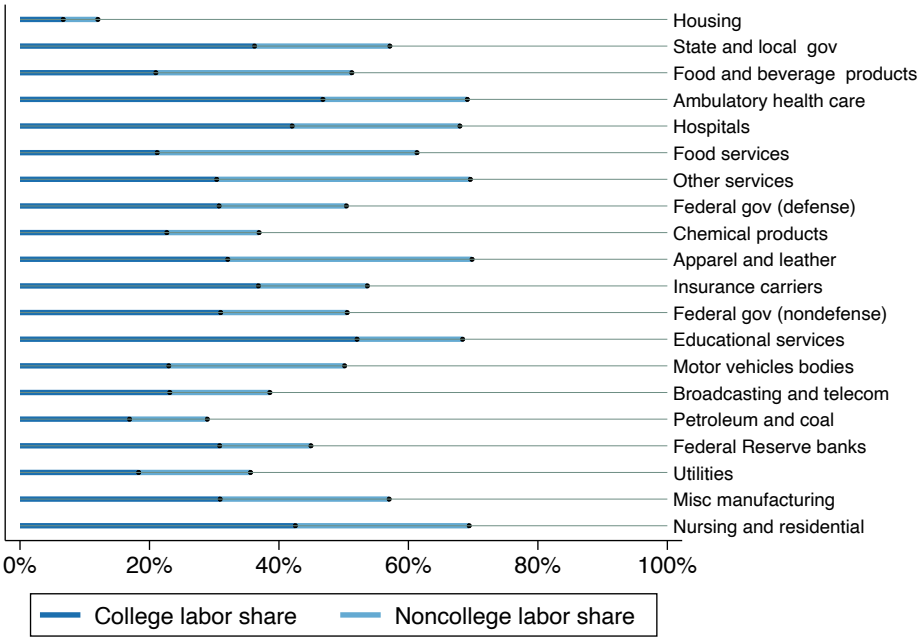
Note that the weighted average labor shares of the consumption and investment sectors sum to the average labor share of the aggregate economy, up to a wedge driven by the difference between labor embodied in exports and imports.

2.3 Results

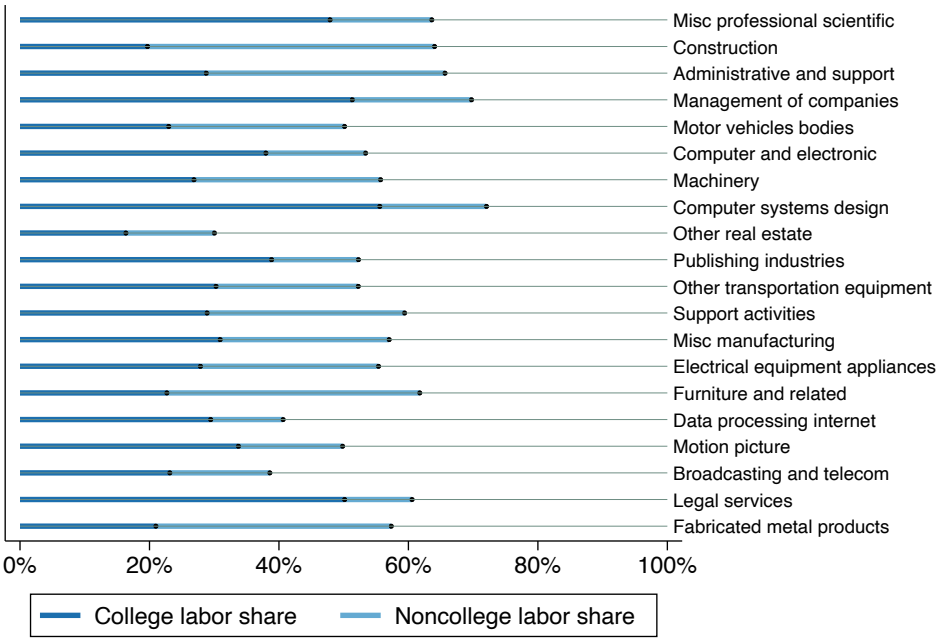
Data from the BEA show that most of the output of each industry is used either for consumption or for investment purposes. Appendix Figure A1 illustrates this pattern: most industries produce primarily either consumption goods or investment goods. This supports our assumption that an industry's income shares are similar regardless of whether its output is used for consumption or investment.

To understand the variation across industries, Figure 3a presents data for the twenty largest industries ranked by their contribution to final consumption expenditures (by households or government). Housing stands out as a major consumption industry and is composed almost entirely of capital services, primarily due to imputed rents from owner-occupied housing. Other significant consumption sectors include food and healthcare.

Figure 3b presents the corresponding data for the twenty largest industries ranked by their contribution to investment expenditures. Industries providing organizational services are par-



(a) Labor share for largest consumption-producing industries



(b) Labor share for largest investment-producing industries

Figure 3: Labor shares by industry in 2017

ticularly important and tend to have high labor shares—especially sectors such as miscellaneous professional and scientific services, administrative and support services, and management of companies. Tangible investment sectors, such as construction, also exhibit high labor shares.

We also report labor shares separately for college- and non-college-educated workers. Consis-

tent with intuition, intangible capital is particularly intensive in college labor, while construction is especially intensive in non-college labor.

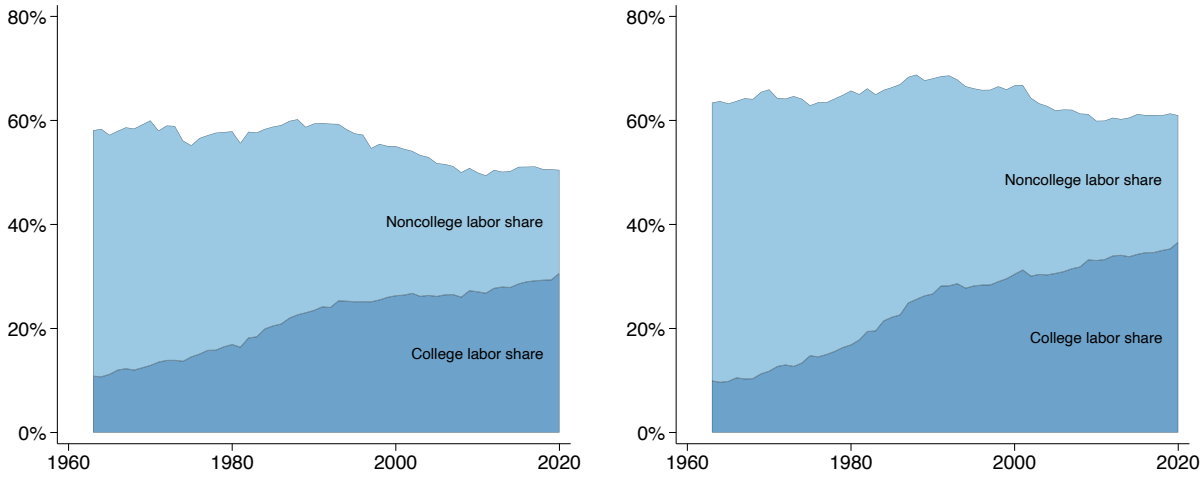
Given the consolidated labor share of each industry, we compute the labor share of the aggregate consumption sector by taking the weighted average of industry labor shares, using the share of final consumption expenditures coming from each industry. We apply the same approach to compute the labor share of the aggregate investment sector:

$$LS_C = \sum_i w_{Ci} LS_i \quad ; \quad LS_I = \sum_i w_{Ii} LS_i \tag{2.1}$$

where w_{Ci} denotes the share of final consumption expenditures in the output of industry i, and w_{Ii} denotes the share of final investment expenditures in the output of industry i.

Figure 4 presents the results. Both sectors begin with a similar labor share in 1962. Over time, however, the labor share in the aggregate consumption sector declines, while the labor share in the investment sector remains relatively stable.

In both sectors, a growing share of labor compensation goes to college-educated workers relative to non-college workers. However, this shift is more pronounced in the investment sector, where the share of payments to college workers has increased more rapidly than in the consumption sector.



(a) Aggregate consumption sector

(b) Aggregate investment sector

Figure 4: Labor income share for consumption and investment sectors

How can we explain the rise in the labor income share of the investment sector? One part of the story is the shift from tangible to intangible investment (Figure 2). Intangible investment is significantly more labor intensive—especially college—labor intensive—than tangible investment (Figure 4). However, another possibility is that all forms of investment have become more labor intensive over time. For instance, automation may have progressed more rapidly in the consumption sector than in the investment sector.

To systematically assess these explanations, we use a shift-share decomposition to analyze the rising difference in labor intensity between the investment and consumption sectors.

Taking the difference of (2.1) over time, denoted Δ, we have:

$$\Delta LS_I - \Delta LS_C = \underbrace{\sum_i (w_{Ii} - w_{Ci})\Delta LS_i}_{\text{Within}} + \underbrace{\sum_i (\Delta w_{Ii} - \Delta w_{Ci})(LS_i + \Delta LS_i)}_{\text{Between}}$$

The first term (the within component) reflects the difference between the weighted average covariance of changes in investment and consumption expenditures and the change in the labor share. It captures the idea that consumption industries may have become more automated over time. The second term (the between component) captures the covariance between the change in sectoral expenditure shares and the labor share. It reflects the shift in the investment sector toward intangible forms of capital, which tend to have a higher labor share.

Table 1 reports the decomposition. We find that roughly half of the rise in labor intensity is accounted for by the within component, and half by the between component. Thus, both channels play a significant role.

Table 1: Decomposing the rise in labor intensity in investment versus consumption sector

	Total	Within	Between
Δ College labor share I vs C	6.9%	5.6%	1.3%
Δ Noncollege labor share I vs C	-1.7%	-2.9%	1.2%
Δ Total labor share I vs C	5.2%	2.7%	2.4%

3 Quantitative model

We now develop and calibrate a more realistic model two-sectors model of the economy based on the data from the national accounts. We use the model to quantify the supply elasticity of capital, both in the short-run and in the long-run. We also emphasize the effect of a higher labor intensity in the investment sector for the incidence of corporate taxation.

3.1 General model

We add four ingredients to the stylized model: (i) multiple types of labor and capital, (ii) type-specific labor supply elasticity, (iii) type-specific internal adjustment costs, and (iv) a tax on capital payouts.

Firm side. The production side of the economy features a consumption good C and N_K differentiated investment goods I_n. These final goods are produced using a combination of N_L labor types

and N_K different capital types.

$$C_t = \prod_{n=1}^{N_K} K_{nC,t}^{\alpha_{nK}} \cdot \prod_{n=1}^{N_L} L_{nC,t}^{\alpha_{nL}} \qquad \text{(consumption good)}$$

$$I_{n,t} = \prod_{n'=1}^{N_K} K_{nI,t}^{\beta_{nn'K}} \cdot \prod_{n'=1}^{N_L} L_{nI,t}^{\beta_{nn'L}} \qquad \text{(investment good)}$$

$$K_{n,t} = K_{nC,t} + K_{nI,t} \qquad \text{(sectoral allocation)}$$

$$K_{n,t+1} = (1-\delta_n)K_{n,t} + K_{n,t}^{\theta_n} I_{n,t}^{1-\theta_n} \qquad \text{(capital accumulation)}$$

In Appendix C.1, we show that this setup is equivalent to an environment where the final goods are produced using intermediary goods, themselves produced using an arbitrary production network. In that case, the relevant cost shares of the value-added function (i.e., the parameters α, β) are known functions of the structure of the network; a feature which we use in the measurement step.

It is helpful to derive the supply-side optimality conditions as the solution to a representative firm who maximizes the present value of payouts net of income taxes:

$$V_0 = \max_{\{L_{nC,t}, L_{nC,t}, K_{nC,t}, K_{nI,t}\}} \sum_{t=1}^{\infty} R_{0\to t}^{-1} D_t$$
$$\text{s.t.} \qquad D_t = (1-\tau_t)\left(C_t - \sum_{n=1}^{N_L} \left(w_{nC,t} L_{nC,t} + w_{nI,t} L_{nI,t}\right)\right)$$

Household side. The household-side is characterized by the optimal supply of (differentiated) labor given relative wages:

$$\frac{L_{nI}}{L_{nC}} = \frac{\mu_n}{1-\mu_n}\left(\frac{w_{nI}}{w_{nC}}\right)^{\phi_n}. \qquad \text{(labor supply)}$$

The total labor supply by labor type n is fixed at $\overline{L}_n$. The share of employment n supplied to each of the final good sectors (C and I) is given by the relative wages and the sectoral labor supply elasticity ϕ_n. When $\phi_n = 0$, it is impossible to reallocate labor from one sector to the other. Conversely, in the case $\phi_n \to \infty$ sectoral labor supply is perfectly elastic, leading to wage equalization between sectors (i.e., $w_{nC} = w_{nI}$) as in the stylized model. This labor supply system, which is identical to what emerges in a Roy-Fréchet selection model, can be microfounded via a CES disutility of labor in a representative household setup.

The labor supply elasticity faced by the investment good sector for labor type n is therefore given by

$$\frac{\mathrm{d}\log L_{nI}}{\mathrm{d}\log w_{nI}} = \left(1 - \frac{L_{nI}}{\overline{L}_n}\right)\phi_n,$$

a fact that we will use to calibrate the parameters ϕ_n.

3.2 Calibration

We now calibrate the model using the cost share data described in the previous section. First, we set the world interest rate to $r = 7\%$, which is approximately the real unlevered return in the US corporate sector over 1970–2020.

Firm side. Our baseline calibration has a single capital type ($N_K = 1$) and two labor types (college and no-college, $N_L = 2$). First, we obtain the production elasticities α (in the consumption sector) and β (in the investment sector). Table 2 reports the baseline calibration, as well as two alternative calibrations (1970 and 2020), which respectively match the 1965–1972 and 2015–2020 periods. We rescale the production elasticities in those two alternatives to maintain the aggregate labor share constant. This allows us to isolate the change in the composition of labor across the two sectors, rather than the overall change in the labor share, which has been widely studied.

To calibrate the capital share in capital formation θ, which governs external adjustment costs, we draw on evidence from Chodorow-Reich, Smith, Zidar and Zwick (2024) on the empirical response of US firms to the 2017 Tax Cuts and Jobs Act. We focus on their result that uses tax files for a sample of roughly $N = 7000$ (nonfinancial, non pass-through domestic firms). The variable of interest is log investment change (post-TCJA versus pre-TCJA). The authors estimate that a 1% decline in the corporate income tax rate leads to a roughly 4% decline in investment (see their Table 3, columns 2 and 3). We interpret their estimates as a partial equilibrium short-run response in the model (i.e., holding constant wages and interest rates). This allows us to identify internal adjustment costs θ separetly from external adjustment costs (i.e., the change in equilibrium wages). In Appendix C.2, we map their evidence to θ directly and obtain $\theta = 0.25$. The key assumption is that firms perceive the tax cut to be permanent.

Household side. The key household-side elasticity is the sectoral supply elasticity ϕ_n and the share parameters μ_n. Together, they govern the how easily labor can be reallocated from the consumption to the investment sector. While the evidence on such elasticities is scarce, the existing literature points to an inelastic supply of scientists and innovators. In the short-run, the stock of scientists is fixed, but even in the medium- and long-run, the high costs and training times imply a low elasticity.

Early evidence from federal r&d spending from Goolsbee (1998) points to an inelastic supply of r&d labor, with an implied elasticity of roughly 0.1 (see David and Hall, 2000). What this means is that demand shocks in the market for r&d workers pass through mostly to higher wages, not more innovation. More recently, Ekerdt and Wu (2024) estimates a rich selection model between r&d and production occupations in the US, which implies a low supply elasticity of roughly 0.3. Using a different approach, Fernández-Villaverde et al. (2024) uses data on the expected payoff of innovation implied the stock market and estimate a supply for the innovative labor of $\phi \approx 0.05$.

To be conservative, we target a labor supply elasticity faced by the investment good sector of

Table 2: General model calibration

Moment	Parameter	Baseline	1970	2020
Consumption sector				
Capital share	α_K	0.36	0.35	0.36
Labor share (no college)	α_{1L}	0.39	0.52	0.27
Labor share (college)	α_{2L}	0.25	0.13	0.37
Investment sector				
Capital share	β_K	0.26	0.28	0.23
Labor share (no college)	β_{1L}	0.46	0.6	0.33
Labor share (college)	β_{2L}	0.28	0.11	0.44
Capital accumulation				
Depreciation	δ	0.15	constant	
External adjustment costs	θ	0.27	constant	
Labor supply				
Share (no college)	μ_1	0.26	constant	
Share (college)	μ_2	0.25	constant	
Elasticity (no college)	ϕ_1	1.35	constant	
Elasticity (college)	ϕ_2	0.4	constant	

0.3, which is the upper bound of the evidence on labor supply elasticity for skilled labor into the innovation sector. For the non college group, we set a higher sectoral elasticity of 1. The reason is that it is presumably easier to transition from construction worker and specialized manufacturing workers, than for a college worker to transition into r&d or management occupations.[5] We use the parameters $(\phi_1, \phi_2, \mu_1, \mu_2)$ to match those labor supply elasticity targets as well as a steady-state wage premium of zero (a normalization).

Capital supply elasticities We now report the model-implied capital supply elasticity. As in the stylized model, we define the capital supply elasticity as the hypothetical response of capital accumulation to a shock to the value of capital q, taking into account the fact that equilibrium wages adjust. In other words, we quantify movements along the supply curve just like in the stylized model. In Appendix C.1, we derive the short- and long-run capital supply elasticities as the solution to matrix equation which contains structural parameters and steady-state shares. Table 3 reports the quantitative results.

The interpretation of the short-run capital supply elasticity is that a hypothetical permanent rise in q of 1% leads to a 0.16% increase in the capital stock within a year, and 1.47% in the long-run. The model therefore implies a realistic response of investment to q. It predicts a low response of capital in the short-run, consistent with the small but positive investment-q correlation in the data, as well as a moderate long-run capital supply elasticity. (See Peters and Taylor, 2017 for

[5]The literature typically finds much higher value for the sectoral labor supply elasticity between occupations with lower educational and training requirements. For instance, Hsieh et al. (2019) estimates an elasticity of 2 across highly-disaggregated (and therefore more similar) occupation groups.

Table 3: Capital supply elasticity.

	General model	Stylized model	NGM	1970 calib.	2020 calib.
Short-run	0.16	9.2	$+\infty$	0.16	0.15
Long-run	1.47	9.2	$+\infty$	1.55	1.4

Notes. The short-run supply elasticity of capital is $\partial \log K_{t+1} / \partial \log q_t$ while the long-run is $\partial \log K / \partial \log q$.

an empirical quantification of the investment-q relationship at the micro and macro level, while accounting for intangible investment.)

To isolate the effect of the additional ingredients introduced in the stylized model (i.e., multiple types of capital and labor and finite sectoral labor supply elasticities), Table 3 reports the capital supply elasticity in the stylized model, calibrated to have the same sectoral labor shares as in the general model, and all other model parameters set to identical values. The stylized model has a much higher capital supply elasticity of 9.2 (roughly 6x larger). Remember that, in the stylized model, the leading term of the capital supply elasticity is one over the excess labor share in the investment sector (see Proposition 1.1). In the calibration, the excess labor share is roughly 0.1 (see Table 2), which explains why the stylized model predicts an elasticity of roughly 10, both in the short- and long-run. The general models predicts a lower long-run elasticity than the stylized model, as well as a more delayed response, driven entirely by external adjustment costs θ_n.

As other benchmark models, we report the elasticities in the NGM (infinite in the short- and long-run) as well as the 1970 and 2020 calibrations. As a result of the relative rise of the labor share in the investment sector over these decades, the implied capital supply elasticities are roughly 10% lower at the end of the sample.

3.3 Quantifying the effect of a corporate tax cut

To illustrate how much this matters for economic outcomes, we now move towards conducting a well-understood model counterfactual: a corporate tax cut. We quantify both the real and redistributional effects, paying attention to how the general model differs from benchmark model and why.

Theory. As discussed in Section 1, a corporate tax cut can be seen as a shift in the demand for capital. As usual in supply-demand systems, a positive demand shock will increase prices and quantities. When the supply curve is horizontal (as in the NGM), then the demand shock is entirely absorbed by quantities K. In the general model, long-run capital supply is imperfectly elastic due to the fact that the investment good is more labor intensive than the consumption good. As a result, a demand shock, in this case a tax cut, leads to a rise of both prices q_t and quantities K_t. To set ideas, consider a permanent decline in τ_K. In Appendix C.2, we use a standard comparative statics approach to obtain expressions for the long-run response of capital, valuations, and wealth

in terms of supply and demand elasticities from the calibrated model (around the undistorted steady-state).

Table 4: Long-run general equilibrium tax elasticities

	general model	stylized model	neoclassical gm	1970 calib.	2020 calib.
Capital stock (K)	0.78	1.24	1.5	0.82	0.75
Capital price (q)	0.54	0.13	0.0	0.53	0.54
Wealth (W)	1.32	1.37	1.5	1.35	1.29

Table 4 reports the resulting long-run tax elasticities. In the calibrated model to calculate the supply and demand elasticities, we obtain that a permanent 1% tax cut leads to a rise in both capital and valuations, in roughly equal parts. In contrast, in the NGM, a same-sized tax cut leads to a rise in capital that is twice as large as in the calibrated model, with no response of valuations.

As we discuss next, this imperfect pass-through of demand shocks to quantities has important effects on the incidence of capital taxes (i.e., who wins and loses in terms of welfare). We first consider an arbitrary perturbation $\{\mathrm{d}\tau_{K,t}\}_{t=0}^{\infty}$ around the undistorted steady-state. Which factors of production win and/or lose in response to this shock? We follow Fagereng et al. (2024) and apply the envelope condition on the household value function to obtain the total welfare effect of the change in prices induced by the tax shock.

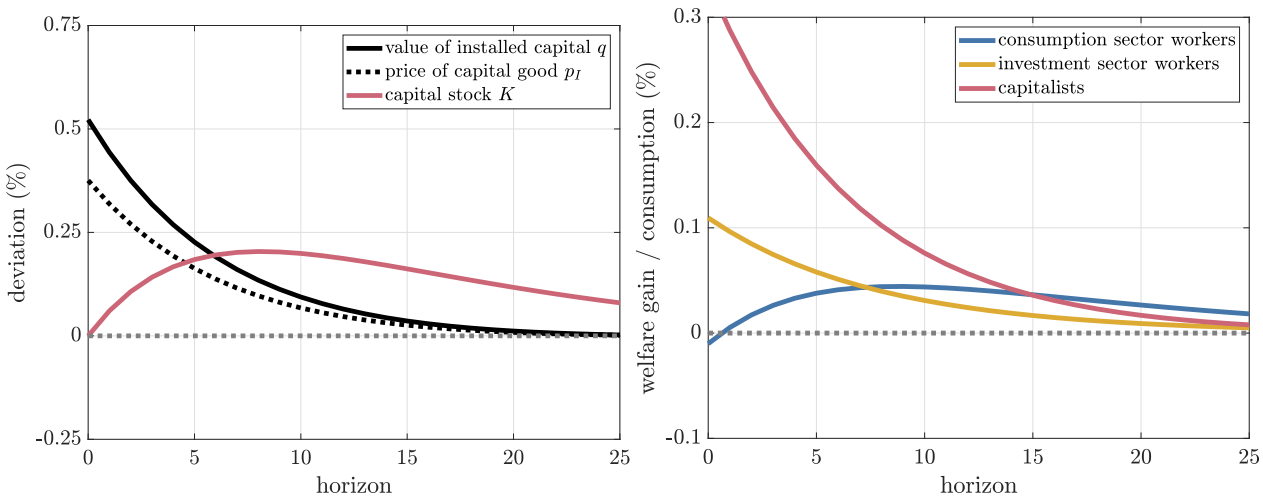
Proposition 3.1 (Tax incidence)**.** *The equilibrium welfare effects, in units of $t = 0$ consumption, associated with the perturbation $\{\mathrm{d}\tau_{K,t}\}_{t=1}^{\infty}$ around the undistorted steady-state, are given by*

$$\text{Welfare Gain}_{ns}^{L} = \sum_{t=1}^{\infty} R^{-t} L_{ns} w_{ns} \cdot \mathrm{d}\log w_{ns,t}$$

$$\text{Welfare Gain}^{K} = \sum_{t=1}^{\infty} R^{-t} D \cdot \mathrm{d}\log(1-\tau_t) - \sum_{n=1}^{N_L} \sum_{s=C,I} \text{Welfare Gain}_{ns}^{L}$$

Proposition (3.1) decomposes the welfare effect on households into the contribution of changes in prices for the three factors of production: labor (by type n and sector s) and capital. For labor, the welfare effect captures the fact that the tax shock affects the path of wages $\{\mathrm{d}w_{nC,t}, \mathrm{d}w_{nI,t}\}_{t=1}^{\infty}$ for each labor type n. Given that capitalists are the residual claimant, we can express their welfare gain as the mechanical effect of the tax cut (i.e., the steady-state capital payout D times the change in the net-of-tax rate $\mathrm{d}\log(1-\tau_t)$) minus the welfare gain of labor.

Quantitative experiment. We now simulate a 1% cut around the steady-state, which decays at an annual rate of 10%. We maintain the small-open economy assumption, where forward returns r are constant. Figure 5a plots the response of capital accumulation and valuations over a 25 year period after the shock. The capital stock peaks at roughly 0.25% higher after 10 years. Both the relative price of capital goods p_I and the value of installed capital q jump on impact.



(a) Capital and valuations

(b) Welfare effects

Figure 5: Model-implied effect of a cut in corporate taxes

Welfare effects. The welfare effect for workers works through changes in their wage, which are themselves fully pinned down by the path of the state and co-state variables $\{q_t, K_t\}_{t=0}^{\infty}$. A rise in capital K benefits both types of workers, via their complementarity with existing capital, but the value of capital q also directly affects the marginal product of investment labor. Figure 5b plots the resulting path of (undiscounted) welfare effects for both types of labor, expressed as a share of steady-state GDP. Notice that capital payout increases are very front-loaded, especially compared to the back-loaded increase in wages (see Figure 5b). This is because consumption-sector wages rise slowly with capital accumulation, while the benefit for capitalists on post-tax capital income is immediate. The wage of investment workers also jumps on impact, reflecting the fact that their marginal product of labor is directly related to the relative price of what they produce (i.e., q).

Table 5 reports the share of private welfare gains that accrue to each factor of production. In the baseline, 11% of the tax incidence falls on consumption-sector workers, while 23% falls on investment-sector labor and 65% on capitalists. Note that this stands in sharp contrast with the NGM, where 100% of the incidence falls on productions workers.

Table 5: Incidence of capital taxes (welfare gain, %).

	General model	Stylized model	NGM	1970 calib.	2020 calib.
Consumption workers	11	38	100	11	12
Investment workers	23	24		23	24
Capitalists	65	38	0	67	65

4 Conclusion

This paper examines how the labor intensity in the investment sector affects the long-run elasticity of capital supply. In our model, producing investment goods requires specialized labor, making capital supply less responsive to changes in asset prices. Unlike the standard neoclassical model, where capital can be frictionlessly accumulated, our framework predicts that the supply of capital becomes inelastic as the share of intangible investment rises.

Using data from the BEA and Compustat, we document that the investment sector has become more labor-intensive, particularly in terms of college-educated workers. This pattern contrasts with the consumption sector, where the labor share has declined. Our findings suggest that the rising labor intensity in investment goods dampens the elasticity of capital supply, leading to higher asset price sensitivity in response to saving shocks.

We then build and calibrate a realistic two-sector economy to quantify the supply elasticity of capital. We find that the long-run supply elasticity of capital is finite, around 1.5, as opposed to the infinite elasticity in the neoclassical model. This lower elasticity has important implications for tax incidence: capital owners and investment workers bear a larger share of the tax burden than production workers. As a result, corporate tax cuts disproportionately benefit those involved in capital formation, exacerbating inequality in intangible-intensive economies.

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Appendix

A Appendix for Section 1

The equilibrium is characterized by

$$
\begin{aligned}
w &= (1-\alpha)(C/L_C) && \text{(labor demand in C)}\\
w &= (1-\beta)(p_I I/L_I) && \text{(labor demand in I)}\\
r_K &= \alpha(C/K_C) && \text{(capital service demand in C)}\\
r_K &= \beta(p_I I/K_I) && \text{(capital service demand in I)}\\
K &= K_C + K_I && \text{(capital market clearing)}\\
\overline{L} &= L_C + L_I && \text{(capital market clearing)}\\
\delta K &= I && \text{(steady-state equation)}\\
q &= \frac{r_K}{r+\delta} && \text{(arbitrage equation)}
\end{aligned}
$$

Steady-state. First we solve for the steady-state sectoral shares (s_K, s_L). Using (capital service demand in I-sector)+(steady-state equation)+(arbitrage equation), we obtain

$$s_K = \beta \frac{\delta}{r+\delta}$$

Then, we combine the four labor/capital demand equations (i.e., using the fact that both sectors face the same relative prices), we have that

$$\frac{\alpha}{\beta}\frac{1-s_L}{s_L} = \frac{1-\alpha}{1-\beta}\frac{1-s_K}{s_K} \implies s_L = \frac{\alpha(1-\beta)s_K}{(\alpha-\beta)s_K + \beta(1-\alpha)}.$$

To obtain the steady-state capital stock, the (steady-state equation) implies

$$K = (\frac{1}{\delta} s_K^{\beta} s_L^{1-\beta})^{\frac{1}{1-\beta}}$$

The resulting capital price can be obtained as

$$q = \frac{\alpha \left(\frac{1-s_L}{1-s_K}\right)^{1-\alpha}}{\beta \left(\frac{s_L}{s_K}\right)^{1-\beta}} K^{\alpha-\beta}.$$

Capital supply elasticity. To isolate the firm-side of the model, we drop the arbitrage equation, which comes from the household-side (i.e., how to price the capital stock K in order to ensure a return r). We focus on first-order dynamics around the steady-state (q, K). We obtain a system of three unknowns (s_K, s_L, K)

and three equations. In terms of log deviations, we have:

$$\begin{aligned}
\alpha\left(\mathrm{d}\log(1-s_K)-\mathrm{d}\log(1-s_L)+\mathrm{d}\log K\right) &= \mathrm{d}\log q+\beta\left(\mathrm{d}\log s_K-\mathrm{d}\log s_L+\mathrm{d}\log K\right),\\
(\alpha-1)\left(\mathrm{d}\log(1-s_K)-\mathrm{d}\log(1-s_L)+\mathrm{d}\log K\right) &= \mathrm{d}\log q+(\beta-1)\left(\mathrm{d}\log s_K-\mathrm{d}\log s_L+\mathrm{d}\log K\right),\\
\mathrm{d}\log K &= \beta\left(\mathrm{d}\log s_K+\mathrm{d}\log K\right)+(1-\beta)\,\mathrm{d}\log s_L.
\end{aligned}$$

The effect of a demand shock (i.e.a movement along the supply curve) can be expressed as the solution to:

$$\begin{pmatrix} -\alpha\frac{s_K}{1-s_K}-\beta & \alpha\frac{s_L}{1-s_L}+\beta & \alpha-\beta\\ -(\alpha-1)\frac{s_K}{1-s_K}-(\beta-1) & (\alpha-1)\frac{s_L}{1-s_L}+(\beta-1) & \alpha-\beta\\ -\beta & -(1-\beta) & 1-\beta \end{pmatrix}\begin{pmatrix}\mathrm{d}\log s_K\\ \mathrm{d}\log s_L\\ \mathrm{d}\log K\end{pmatrix}=\begin{pmatrix}1\\1\\0\end{pmatrix}\mathrm{d}\log q$$

The solution is

$$\begin{pmatrix}\mathrm{d}\log s_K\\ \mathrm{d}\log s_L\\ \mathrm{d}\log K\end{pmatrix}=\frac{1}{\alpha-\beta}\begin{pmatrix}1-\beta\\ \frac{1-s_L}{1-s_K}(1-\beta)\\ \frac{1-(\beta s_K+(1-\beta)s_L)}{1-s_K}\end{pmatrix}\mathrm{d}\log q$$

The capital supply elasticity is therefore given by

$$\begin{aligned}
\frac{\partial\log K^s}{\partial\log q} &= \frac{1}{\alpha-\beta}\cdot\frac{1-(\beta s_K+(1-\beta)s_L)}{1-s_K},\\
&= \frac{1}{\alpha-\beta}-\frac{(1-\beta)\delta}{r(1-\alpha)+(1-\beta)\delta},
\end{aligned}$$

where the second equality uses the steady-state formulas for s_L and s_K.

Capital demand elasticity. Using the arbitrage equation, we have that $\partial\log q=\partial\log r_K$. Using (capital service demand in C-sector), we have that

$$\begin{aligned}
1 &= (\alpha-1)\frac{\partial\log K^d}{\partial\log q}+(\alpha-1)\left(\frac{s_L}{1-s_L}\frac{\partial\log s_L}{\partial\log q}-\frac{s_K}{1-s_K}\frac{\partial\log s_K}{\partial\log q}\right)\\
\frac{\partial\log K^d}{\partial\log q} &= -\left(\frac{1}{1-\alpha}+\frac{s_L}{1-s_L}\frac{\partial\log s_L}{\partial\log q}-\frac{s_K}{1-s_K}\frac{\partial\log s_K}{\partial\log q}\right).
\end{aligned}$$

Plugging-in the reallocation effect (i.e., $\frac{\partial\log s_L}{\partial\log q}$ and $\frac{\partial\log s_L}{\partial\log q}$ calculated in as part of the capital supply elasticity), we obtain the capital demand elasticity:

$$\begin{aligned}
\frac{\partial\log K^d}{\partial\log q} &= -\left(\frac{1}{1-\alpha}+\frac{1-\beta}{\alpha-\beta}\frac{s_L-s_K}{1-s_K}\right),\\
&= -\left(\frac{1}{1-\alpha}+\frac{(1-\beta)\delta}{r(1-\alpha)+(1-\beta)\delta}\right),
\end{aligned}$$

where the second equality uses the steady-state formulas for s_L and s_K.

Case study: capital income tax cut. Consider a shock $\mathrm{d}\log(1-\tau)$ in the arbitrage equation, where τ is the capital income tax rate. What is the effect on the equilibrium (q, K)?

$$\mathrm{d}\log K = \frac{\partial \log K^s}{\partial \log q} \cdot \mathrm{d}\log q \tag{supply}$$

$$\mathrm{d}\log K = \frac{\partial \log K^d}{\partial \log q} \cdot \mathrm{d}\log q + \frac{1}{1-\alpha}\,\mathrm{d}\log(1-\tau) \tag{demand}$$

The price response is[6]

$$\begin{aligned}\frac{\mathrm{d}\log q}{\mathrm{d}\log(r+\delta)} &= \frac{\frac{1}{1-\alpha}}{\frac{\partial \log K^s}{\partial \log q} - \frac{\partial \log K^d}{\partial \log q}} \\ &= \frac{\alpha-\beta}{1-\beta}.\end{aligned}$$

The quantity response is

$$\begin{aligned}\frac{\mathrm{d}\log K}{\mathrm{d}\,\mathrm{d}\log(1-\tau)} &= \frac{\alpha-\beta}{1-\beta}\frac{\partial \log K^s}{\partial \log q} \\ &= \frac{1}{1-\beta} - \frac{(\alpha-\beta)\delta}{(1-\alpha)r + (1-\beta)\delta}.\end{aligned}$$

The sign of the price response depends on $\alpha - \beta$, but the quantity response is always negative.

B Appendix for Section 2

B.1 Evidence from national accounts

Table A2 lists all industries included in our empirical analysis. For each industry, we report three categories: (i) the value of the industry's final use as a share of GDP; (ii) the composition of this final use, broken down into consumption, tangible investment, intangible investment, and net exports (exports minus imports); and (iii) the consolidated labor share of the industry's output, which accounts for the labor embodied in all intermediate inputs used by the industry.

Figure A1 visualizes one component of Table A2: the share of each industry's final use that is allocated to investment rather than consumption. The key takeaway from this figure is that most industries are used *either* for consumption or for investment, but not both. This is reassuring, as our calculation of the labor share for the aggregate consumption and investment sectors relies on the assumption that an industry's labor share is the same whether its output is used for consumption or investment. Since most industries are predominantly allocated to one category, this assumption is less restrictive in practice. The issue becomes even less significant when using the detailed input-output table with 402 industries, where an even larger fraction of industries is used primarily for either consumption or investment.

[6] We can also conduct the equilibrium comparative statics calculation using Baqaee and Malmberg (2025). Denoting $\Psi = (I - \Omega)^{-1}$ the Leontief inverse of the cost share matrix, we could have also obtained as

$$\frac{\mathrm{d}\log q}{\mathrm{d}\log(1-\tau)} = \Psi_{IL} - 1 - \Psi_{CL} = \frac{\alpha-\beta}{1-\beta}.$$

Table A1: Full list of use shares and income shares by industry (2017)

Industry	Uses/GDP	Final use shares				Consolidated labor shares	
		C	I_{tan}	I_{intan}	X − M	College	Noncollege
Farms	1%	72%	0%	0%	28%	20%	35%
Forestry fishing	0%	50%	0%	0%	50%	20%	54%
Mining except oil	0%	2%	1%	0%	98%	17%	24%
Support activities	0%	0%	97%	0%	3%	29%	31%
Utilities	1%	99%	0%	0%	1%	18%	17%
Construction	6%	0%	100%	0%	0%	20%	44%
Wood products	0%	28%	42%	0%	30%	19%	40%
Nonmetallic mineral products	0%	64%	0%	0%	36%	19%	29%
Primary metals	0%	4%	0%	0%	96%	19%	31%
Fabricated metal products	0%	41%	19%	0%	40%	21%	36%
Machinery	2%	6%	68%	0%	26%	27%	29%
Computer and electronic	3%	25%	55%	0%	20%	38%	15%
Electrical equipment appliances	1%	47%	31%	0%	22%	28%	28%
Motor vehicles bodies	3%	43%	41%	0%	16%	23%	27%
Other transportation equipment	1%	14%	43%	0%	44%	30%	22%
Furniture and related	1%	67%	29%	0%	3%	23%	39%
Misc manufacturing	1%	68%	19%	0%	13%	31%	26%
Food and beverage products	5%	93%	0%	0%	7%	21%	30%
Textile mills	0%	77%	11%	0%	12%	24%	32%
Apparel and leather	2%	98%	0%	0%	2%	32%	38%
Paper products	0%	70%	0%	0%	30%	22%	30%
Printing and related	0%	77%	0%	0%	23%	24%	34%
Petroleum and coal	2%	78%	0%	0%	22%	17%	12%
Chemical products	3%	74%	0%	0%	26%	23%	14%
Plastics and rubber	0%	69%	1%	0%	30%	23%	27%
Other retail	0%	100%	0%	0%	0%	27%	35%
Air transportation	1%	75%	0%	0%	25%	27%	22%
Rail transportation	0%	52%	0%	0%	48%	18%	31%
Water transportation	0%	82%	0%	0%	18%	24%	24%
Truck transportation	0%	88%	0%	0%	12%	16%	43%
Transit and ground	0%	100%	0%	0%	0%	22%	36%
Other transportation	0%	64%	0%	0%	36%	24%	40%
Warehousing and storage	0%	60%	0%	0%	40%	18%	46%
Publishing industries	1%	44%	0%	43%	13%	39%	13%
Motion picture	0%	32%	0%	42%	26%	34%	16%
Broadcasting and telecom	2%	87%	2%	8%	3%	23%	15%
Data processing internet	0%	48%	0%	46%	6%	29%	11%
Federal Reserve banks	1%	82%	0%	0%	18%	31%	14%
Securities	1%	76%	0%	0%	24%	57%	12%
Insurance carriers	2%	94%	1%	0%	5%	37%	17%
Funds trusts	1%	100%	0%	0%	0%	48%	13%
Housing	8%	100%	0%	0%	0%	7%	5%
Other real estate	1%	3%	95%	0%	2%	16%	14%
Rental and leasing	1%	55%	0%	0%	45%	17%	15%
Legal services	1%	74%	16%	0%	10%	50%	10%
Computer systems design	1%	0%	13%	78%	9%	56%	16%
Misc professional scientific	8%	4%	3%	85%	8%	48%	16%
Management of companies	2%	0%	0%	99%	1%	51%	18%
Administrative and support	3%	9%	0%	91%	0%	29%	37%
Waste management	0%	99%	0%	0%	1%	20%	35%
Educational services	2%	99%	0%	0%	1%	52%	16%
Ambulatory health care	4%	100%	0%	0%	0%	47%	22%
Hospitals	4%	100%	0%	0%	0%	42%	26%
Nursing and residential	1%	100%	0%	0%	0%	43%	27%
Social assistance	1%	100%	0%	0%	0%	38%	35%
Performing arts spectator	0%	93%	0%	6%	2%	44%	24%
Amusements gambling	1%	100%	0%	0%	0%	28%	32%
Accommodation	1%	100%	0%	0%	0%	24%	31%
Food services	3%	100%	0%	0%	0%	21%	40%
Other services	3%	100%	0%	0%	0%	30%	39%
Federal gov (defense)	2%	100%	0%	0%	0%	31%	20%
Federal gov (nondefense)	2%	100%	0%	0%	0%	31%	20%
Federal gov enterprises	0%	91%	0%	0%	9%	30%	20%
State and local gov	7%	100%	0%	0%	0%	36%	21%
State and local gov enterprises	0%	100%	0%	0%	0%	32%	23%

Table A2 compares the labor share computed using our baseline industry classification with the version based on the more detailed 402-industry classification for the year 2017. The results are very similar, indicating that using a coarser industry breakdown—necessary to extend the analysis back to 1963—does not materially affect our findings.

B.2 Evidence from Compustat

In the main text, we use data from the BEA to compute the labor share in the consumption and investment sectors. We now show that our results remain similar if we use firm-level accounting data from Compustat

Figure A1: Investment as fraction of final use by industry

Table A2: Robustness of labor income shares

	College labor share	Noncollege labor share
Panel A: Consumption sector		
Detailed	28.2%	21.7%
Summary	29.2%	22.0%
Panel B: Investment sector		
Detailed	33.2%	23.6%
Summary	34.6%	26.5%

instead (e.g., Peters and Taylor, 2017, Eisfeldt and Papanikolaou, 2013, He et al., 2024).

B.2.1 Methodology

We use Compustat-CRSP merged data covering the period from 1972–2022. We use usual screens to focus on the nonfinancial corporate sector. Our goal is to construct industry-level series for tangible investments,

investment in R&D, and investment in organizational services.

We start from the statement of cashflows, as reported in Compustat. The key accounting identity is the cashflows from operations, the cashflows from financing activities, and cashflows for investing activities sum up to zero. We use the convention that changes in cash balances represent net payments to debt holders. Notice that this cashflow identity is the financial accounting counterpart of the firm budget constraint in the model. These equations allow us to account for 100% of the cash that comes in due to profits (cashflows from operations), the net cash that comes out to pay owners (cashflows from financing activities), and the cash that comes out due to investment (cashflows from investment activities).

Adjustment #1: expensed payments to investment labor. One issue with accounting data is that the distinction between an expense (which are subtracted from sales to obtain cashflows from operation) and investment (which are not) can be arbitrary, especially in the case of intangibles.

The existing literature shows that this issue leads to an understatement of cashflows from operations and a corresponding understatement of (minus) cashflows from investment (e.g., Peters and Taylor (2017) for evidence from firm-level data and Koh et al. (2020) from aggregate data). A usual rule of thumb in the literature is to reallocate 30% of SG&A and 100% of R&D from expenses to intangible capital expenditures. This is meant to account for the incorrect expensing of labor expenses that contributes to organizational capital production. Recently, He et al. (2024) shows that salaries in sales and marketing alone account for

$$\text{cf adjustment (investment labor)} \equiv 0.3 \cdot \text{SG\&A} + \text{R\&D}.$$

Adjustment #2: net entry in sample. When a private firm becomes public or goes public, it effectively represents a negative cashflow for passive capitalists, who re-balance their portfolio to always own the market. Similarly, when a firm exits or gets acquired, it represents a positive cashflow. We now describe how we measure the contribution of net firm entry to market capitalization growth.

First, we introduce a decomposition of the total market value growth (at the aggregate or within an industry) between time t and $t+1$ as the sum of three components due to stayers, entrants, and exits. Formally, we have that

$$g_{\text{total}} = g_{\text{stayer}} + g_{\text{entry}} + g_{\text{exit}}, \tag{B.1}$$

where the components are defined in this footnote.[7]

Using these definitions, we define the net cashflow due to net firm entry as

$$\text{cf adjustment (net entry)} \equiv -(g_{\text{entry}} + g_{\text{exit}}) \cdot \text{market capitalization}. \tag{B.2}$$

It corresponds to the net cash that a passive capitalists who owns the market receives due to net firm entry during a period.

[7] The formulas are:

$$g_{\text{total}} = \frac{\sum_{i\in\mathcal{F}_{t+1}} V_{i,t+1}}{\sum_{i\in\mathcal{F}_t} V_{i,t}} - 1; \qquad g_{\text{stayer}} = \frac{\sum_{i\in(\mathcal{F}_t\cap\mathcal{F}_{t+1})} V_{i,t+1}}{\sum_{i\in(\mathcal{F}_t\cap\mathcal{F}_{t+1})} V_{i,t}} - 1,$$

$$g_{\text{entry}} = \frac{\sum_{i\in(\mathcal{F}_{t+1}\setminus\mathcal{F}_t)} V_{i,t+1}}{\sum_{i\in\mathcal{F}_t} V_{i,t}}; \qquad g_{\text{exit}} = -\frac{(1+g_{\text{stayer}})\sum_{i\in(\mathcal{F}_t\setminus\mathcal{F}_{t+1})} V_{i,t+1}}{\sum_{i\in\mathcal{F}_t} V_{i,t}},$$

where $\mathcal{F}_t$ is the universe of firms at time t and V_i is the market capitalization of firm i.

Adjustment #3: payments in stocks. How should we record a payment when it is the form of stocks? First, we can decompose the growth in market value of a stayer into growth of price per share and the growth in number of shares. To match the timing of the payment to the timing of the stock issuance, we use the fully-diluted share count (which includes outstanding shares, as well as all possible sources of potential shares such as stock options and reserved shares).

Formally, we can decompose the growth of the market capitalization of stayers (i.e., firms who do not exit the sample) defined in (B.1) into a price and number of shares component:

$$g_{\text{stayer}} = g_{\text{price}} + g_{\text{shares}},$$

where the components are defined in this footnote.[8]

Moreover, we observe the contribution of the growth in the number of shares issued for cash $g_{\text{shares, cash}}$, and the remainder $g_{\text{shares}} - g_{\text{shares, cash}}$ is due to a combination of stock compensation and stock-financed acquisitions (i.e., "noncash payments"). Using the same logic as for firm entry (see equation B.2), we define noncash payments as

$$\text{cf adjustment (noncash payments)} \equiv (g_{\text{shares}} - g_{\text{shares, cash}}) \cdot \text{market capitalization}.$$

The breakdown between stock compensation and acquisitions is only available after 2011. Therefore, we assume that acquisition is a constant share $\omega \in (0, 1)$ of noncash payments (stock issuance not associated with cash). We compute ω at the industry level after 2011, and use it to split stock compensation before 2011.

Adjustment #4: imputation of the tangible share of ambiguous investments. How should one proceed to allocate the sum of cashflows due to mergers, acquisitions, and IPOs into tangible versus intangible investment? To do so, we use the "ambiguous income" approach introduced in Cooley et al. (1995). In our setup, the idea will be to assume that acquisitions have the same intangible content as other forms of investment. Denoting $\widetilde{\chi} \in (0, 1)$ the intangibility of investment in the rest of investment, we have that

$$\widetilde{\chi} \equiv \frac{\text{cf adjustments}\left(\text{investment labor} + \psi \cdot (1-\omega) \cdot \text{noncash payments}\right)}{\text{cf from investing (capex)} + \text{cf adjustments}\left(\text{investment labor} + \psi \cdot (1-\omega) \cdot \text{noncash payments}\right)},$$

where we discuss the role of ψ shortly. We compute $\widetilde{\chi}$ at the industry year-level. Equipped with this last estimate, we are ready to construct our final variables.

[8] The definitions of g_{stayer}, g_{price}, and g_{shares} are

$$g_{\text{stayer}} = \frac{P_{i,t+1} N_{i,t+1}}{P_{i,t} N_{i,t}} - 1; \qquad g_{\text{price}} = \frac{P_{i,t+1}}{P_{i,t}} - 1; \qquad g_{\text{shares}} = \left(\frac{N_{i,t+1}}{N_{i,t}} - 1\right)(1 + g_{\text{price}}),$$

where $P_{i,t}$ is the price per share and $N_{i,t}$ is the number of shares outstanding at firm i period t.

Resulting formulas. Having defined the key variables, we now write down the final formulas:

$$\begin{aligned}
\text{tangible investments} = &- \text{cf from investing (capex)},\\
&- (1-\widetilde{\chi}) \cdot \text{cf from investing (acquisition)},\\
&- (1-\widetilde{\chi}) \cdot \text{cf adjustments} \Big(\text{net entry} + \omega \cdot \text{noncash payments}\Big),\\
\text{intangible investments} = &- \text{cf adjustments} \Big(\text{investment labor} + \psi \cdot (1-\omega) \cdot \text{noncash payments}\Big),\\
&- \widetilde{\chi} \cdot \text{cf from investing (acquisition)},\\
&- \widetilde{\chi} \cdot \text{cf adjustments} \Big(\text{net entry} + \omega \cdot \text{noncash payments}\Big),\\
\text{payments to capitalists} = &- \text{cf from financing}\\
&+ \text{cf adjustments} \Big(\text{net entry} + \text{noncash payments}\Big).
\end{aligned}$$

Combining these formulas, we obtain an expression for capital income

$$\begin{aligned}
\text{capital income} =& \text{tangible investments} + \text{intangible investments} + \text{payments to capitalists}\\
=& \text{cf from operations} + \text{cf adjustments (investment labor)}\\
&- (1-\psi) \cdot (1-\omega) \cdot \text{noncash payments},
\end{aligned}$$

where the second equality uses the accounting identity of the cashflow statement. The number $\psi(1-\omega)$ corresponds to the share of noncash payments that is unreported in R&D and SG&A. Following changes in accounting standards in 2006, we set $\psi = 0.5$ before 2006 and $\psi = 0$ afterwards.

A few remarks are in order. First, that payments to capitalists are invariant to the calibration $(\widetilde{\chi}, \omega)$. This is because they only represent assumptions on how cashflows are distributed between tangible investments and intangible investments. Second, capital income is invariant not only to the calibration $(\widetilde{\chi}, \omega)$, but also the adjustments for noncash payments and net entry. This is because noncash payments and net entry are purely redistributive flows (from capitalists to tangible investments and to innovators). The only adjustment that affects the level of capital income is the adjustment for expensed investment labor. The idea is that we reclassify line items as capital expenditures, not expenses.

B.2.2 Results

Figure A2a shows the distribution of investment as a proportion of value added over time. Three patterns stand out. First, the share of tangible investment declines. Second, the share of intangible investment rises, driven by increases in both R&D and organization capital—as proxied by the SG&A component in Compustat. Third, this shift reflects a broader rise in investment intangibility: from around 10% in the 1970s to 20% after 2000.

Table A3 reports the average distribution of capital income over the 1972–2022 period, including a detailed breakdown, and robustness checks. Tangible investments account for 42% of capital over the period, mostly due to direct capital expenditures, but also in part due to acquisition of existing businesses (who themselves are partial tangible). Intangible investments account for 44%. Cash and noncash compensation to investment labor accounts for most of it, but almost 20% comes from payouts associated with cash and noncash payout that arise in acquisitions. The most important line item is the component of sg&a that we

attribute to management, marketing, and management labor (see He, Mostrom and Sufi (2024) for recent evidence).

Finally, payments to capitalists account for a mere 14% of capital income over the sample. While net cash payouts to equity holders account for 19% of capital income, they were offset by a 12% equity dilution (i.e., noncash payments). This sample average is skewed by the tech bubble of the 1990s, where payments to capitalists were negative for a few years (i.e., acquisitions and IPOs exceeded dividends and interest payouts, see Fried and Wang, 2019 for a discussion of this fact).

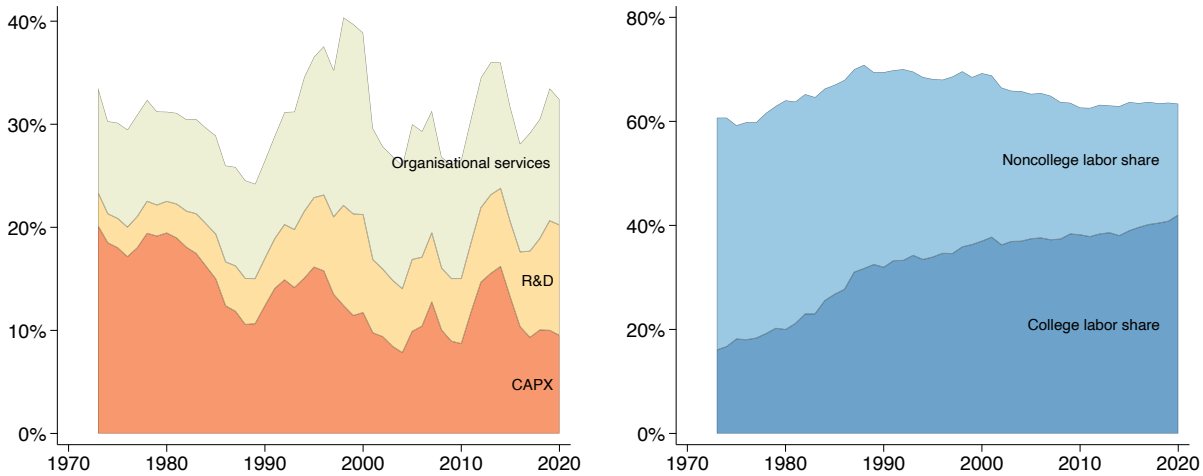
Table A3: Distribution of capital income (1972–2022 average).

Uses of capital income (%)	Baseline	Robustness			
		$\tilde{\chi} = 0$	$\tilde{\chi} = 1$	$\omega = 0$	$\omega = 1$
Tangible investments	36	45	37	37	42
Tangible capital expenditures	37	37	37	37	37
Mergers, acquisitions, and IPOs	-1	8	0	0	5
$(1-\tilde{\chi}) \cdot$ cash acquisitions	3	8	0	3	3
$(1-\tilde{\chi})(1-\omega) \cdot$ non-cash payments	2	5	0	0	5
$(1-\tilde{\chi}) \cdot$ net entry in public universe	-6	-6	0	-3	-3
Intangible investments	46	38	46	42	46
Intangible capital expenditure	38	38	38	39	35
$0.3 \cdot$ selling, general, and admin. expenses	24	24	24	24	24
research and development expenses	12	12	12	12	12
$\psi \cdot \omega \cdot$ non-cash payments	2	2	2	4	0
Mergers, acquisitions, and IPOs	8	0	8	3	10
$\tilde{\chi} \cdot$ cash acquisitions	5	0	8	5	5
$\tilde{\chi}(1-\omega) \cdot$ non-cash payments	3	0	5	0	7
$\tilde{\chi} \cdot$ net entry in public universe	1	0	-6	-2	-2
Payout to capitalists	18	18	18	18	18
Net cash equity payout	19	19	19	19	19
− Non-cash payments	-13	-13	-13	-13	-13
− Net entry in public universe	6	6	6	6	6
Net debt payout	6	6	6	6	6

To explore the implications of these changing investment expenditures on the labor component of investment, we combine these investment shares with data on the labor component of tangible investment (equipment and structures), R&D, and organization capital from the BEA. We then construct the labor share of investment as the weighted average labor share across these three components. Note that both the weights and the labor shares within each investment type change over time. The resulting labor share, shown in Figure 4b, evolves similarly to the BEA series. Notably, we find that the labor share of investment increases—a contrast to the well-documented decline in the aggregate labor share of the economy.

Specifically, Figure A2a shows a large rise in the intangibility of investment, from roughly 10% in the 1970s to 20% post-2000. We then combine these changing investment expenditures with the labor component of tangible investment (equipment and structure), R&D, and Organization capital from the BEA. The result, plotted in Figure 4b show a similar evolution of the labor share relative to the BEA. Actually, we find that the labor share of investment *increases*, which contrasts with the well-known decline in the labor share in the aggregate economy.

Overall, these results show that results form the BEA and Compustat show similar trends regarding the rising importance of intangibles in investment, and, as a result, the reliance of capital formation on high-skill labor.



(a) Composition of investment

(b) Labor share of investment sector

Figure A2: Labor income share for consumption and investment sectors

C Appendix for Section 3

C.1 Equilibrium of general model

We now focus on the case $N_K = 1$, $N_L = 2$ and drop the time subscripts. The Lagrangian for the firm problem is

$$\begin{aligned}
\mathcal{L}_0 = & \sum_{t=1}^{\infty} R_{0\to t}^{-1} \left((1-\tau_t) \left(\prod_{n=1}^{N_K} K_{nC,t}^{\alpha_{nK}} \cdot \prod_{n=1}^{N_L} L_{nC,t}^{\alpha_{nL}} - \sum_{n=1}^{N_L} \left(w_{nC,t} L_{nC,t} + w_{nI,t} L_{nI,t} \right) \right) - \tau_t p_{I,t} I_t \right) \\
& + \sum_{t=1}^{\infty} R_{0\to t}^{-1} r_{K,t} \left(K_{C,t} + K_{I,t} - K_t \right) \\
& + \sum_{t=1}^{\infty} R_{0\to t}^{-1} p_{I,t} \left(I_{n,t} - I_t \right) \\
& + \sum_{t=1}^{\infty} R_{0\to t}^{-1} q_t \left((1-\delta) K_t + K_t^{\theta} I_t^{1-\theta} - K_{t+1} \right)
\end{aligned}$$

Notice that, unlike in the stylized model, there is distinction between the price (i.e., shadow value) of installed capital q_t versus capital goods $p_{I,t}$. This is exactly as in the q-theory of investment, where, in the short-run, there is a wedge between the price of investment good and the implied value via asset prices. In our case, we can express the value of the firm $V_t = q_t K_{t+1}$ (i.e., the value of the whole corporate sector).

The equilibrium conditions are

$$\alpha_{1L}C = w_{1C}L_{1C}, \quad \alpha_{2L}C = w_{2C}L_{2C}, \quad \alpha_K C = \frac{w_K}{1-\tau}K_C \tag{focs - consumption good}$$

$$\beta_{1L}p_I I = w_{1I}L_{1I}, \quad \beta_{2L}p_I I = w_{2I}L_{2I}, \quad \beta_K p_I I = \frac{w_K}{1-\tau}K_I \tag{focs - capital good}$$

$$\frac{L_{1C}}{L_{1I}} = \frac{1-\mu_1}{\mu_1}\left(\frac{w_{1C}}{w_{1I}}\right)^{\phi}, \quad \frac{L_{2C}}{L_{2I}} = \frac{1-\mu_2}{\mu_2}\left(\frac{w_{2C}}{w_{2I}}\right)^{\phi} \tag{focs - sorting}$$

$$L_{1C} + L_{1I} = L_1, \quad L_{2I} + L_{2C} = L_2, \quad K_C + K_I = K \tag{market clearing}$$

$$K' = (1-\delta)K + K^{\theta}I^{1-\theta} \tag{capital accumulation}$$

$$\frac{I}{K} = (1-\theta)^{\frac{1}{\theta}}\left(\frac{q}{p_I}\right)^{\frac{1}{\theta}} \tag{investment-q}$$

$$(1+r')q = r'_K + \left(1-\delta+\theta\left(\frac{I'}{K'}\right)^{1-\theta}\right)q' \tag{arbitrage equation}$$

Steady-state. We now characterize the undistorted steady-state of the model (with $\tau = 0$)

$$\alpha_{1L}C = w_{1C}L_{1C}, \quad \alpha_{2L}C = w_{2C}L_{2C}, \quad \alpha_K C = \frac{w_K}{1-\tau}K_C \tag{focs - consumption good}$$

$$\beta_{1L}p_I I = w_{1I}L_{1I}, \quad \beta_{2L}p_I I = w_{2I}L_{2I}, \quad \beta_K p_I I = \frac{w_K}{1-\tau}K_I \tag{focs - capital good}$$

$$\frac{L_{1I}}{L_{1C}} = \frac{\mu_1}{1-\mu_1}\left(\frac{w_{1I}}{w_{1C}}\right)^{\phi}, \quad \frac{L_{2I}}{L_{2C}} = \frac{\mu_2}{1-\mu_2}\left(\frac{w_{2I}}{w_{2C}}\right)^{\phi} \tag{focs - sorting}$$

$$L_{1C} + L_{1I} = L_1, \quad L_{2I} + L_{2C} = L_2, \quad K_C + K_I = K \tag{market clearing}$$

$$\frac{I}{K} = (1-\theta)^{\frac{1}{\theta}}\left(\frac{q}{p_I}\right)^{\frac{1}{\theta}} \tag{investment-q}$$

$$\delta^{\frac{1}{1-\theta}}K = I, \quad q = \frac{r_K}{r+(1-\theta)\delta} \tag{steady-state + arbitrage}$$

First, using (investment-q) and (steady-state), we have that the relative price of the capital versus investment goods is

$$\frac{q}{p_I} = \frac{\delta^{\frac{\theta}{1-\theta}}}{1-\theta}$$

Then, we obtain the long-run share of capital allocated to the investment sector is $s_K \equiv K_I/K$ using (focs - capital good - 3) and (steady-state + arbitrage):

$$s_K = \beta_K \frac{(1-\theta)\delta}{r+(1-\theta)\delta}$$

Plugging into (focs - consumption good - 3 + focs - capital good - 3), we have that consumption share of GDP is

$$\frac{C}{p_I I} = \frac{\beta_K}{\alpha_K}\frac{1-s_K}{s_K}$$

To obtain the employment shares s_{nL}, we use(focs - sorting):

$$\frac{1-s_{nL}}{s_{nL}} = \frac{1-\mu_n}{\mu_n}\left(\frac{\alpha_{nL}/\alpha_K}{\beta_{nL}/\beta_K}\frac{s_{nL}}{1-s_{nL}}\frac{1-s_K}{s_K}\right)^{\phi}$$
$$\implies \frac{1-s_{nL}}{s_{nL}} = \left(\frac{1-\mu_n}{\mu_n}\right)^{\frac{1}{1+\phi}}\left(\frac{\alpha_{nL}/\alpha_K}{\beta_{nL}/\beta_K}\frac{1-s_K}{s_K}\right)^{\frac{\phi}{1+\phi}}$$

Equipped with (s_K, s_{1L}, s_{2L}), it is trivial to solve the rest of the steady-state allocation (q, K).

Capital supply elasticity. From now on, we drop the arbitrage equation and have q be exogeneous. This allows us to analyze the supply-side of the economy. In the short-run, K is fixed, so that there are only 4 unknowns $(s_K, s_{1L}, s_{2,L}, p_I)$ and 4 equations:

$$\begin{aligned}
\frac{\alpha_K C}{\beta_K p_I I} &= \frac{K_C}{K_I}, \\
\frac{\alpha_{1L} C}{\beta_{1L} p_I I} &= \left(\frac{\mu_1}{1-\mu_1}\right)^{\frac{1}{\phi}}\left(\frac{L_{1C}}{L_{1I}}\right)^{1+\frac{1}{\phi}}, \\
\frac{\alpha_{2L} C}{\beta_{2L} p_I I} &= \left(\frac{\mu_2}{1-\mu_2}\right)^{\frac{1}{\phi}}\left(\frac{L_{2C}}{L_{2I}}\right)^{1+\frac{1}{\phi}}, \\
\frac{I}{K} &= (1-\theta)^{\frac{1}{\theta}}\left(\frac{q}{p_I}\right)^{\frac{1}{\theta}}
\end{aligned}$$

The short-run first-order dynamics are summarized by an equation of the form $Ax = b$, where

$$A(\theta) = \begin{pmatrix}
\frac{s_K}{1-s_K}(\alpha_K - 1) + (\beta_K - 1) & \frac{s_{1L}}{1-s_{1L}}\alpha_{1L} + \beta_{1L} & \frac{s_{2L}}{1-s_{2L}}\alpha_{2L} + \beta_{2L} & 1 \\
\frac{s_K}{1-s_K}\alpha_K + \beta_K & \frac{s_{1L}}{1-s_{1L}}\left(\alpha_{1L} - \frac{1+\phi}{\phi}\right) + \left(\beta_{1L} - \frac{1+\phi}{\phi}\right) & \frac{s_{2L}}{1-s_{2L}}\alpha_{2L} + \beta_{2L} & 1 \\
\frac{s_K}{1-s_K}\alpha_K + \beta_K & \frac{s_{1L}}{1-s_{1L}}\alpha_{1L} + \beta_{1L} & \frac{s_{2L}}{1-s_{2L}}\left(\alpha_{2L} - \frac{1+\phi}{\phi}\right) + \left(\beta_{2L} - \frac{1+\phi}{\phi}\right) & 1 \\
\theta\beta_K & \theta\beta_{1L} & \theta\beta_{1L} & 1
\end{pmatrix}$$
$$x = \begin{pmatrix} \frac{\mathrm{d}\log s_K}{\mathrm{d}\log q} & \frac{\mathrm{d}\log s_{1L}}{\mathrm{d}\log q} & \frac{\mathrm{d}\log s_{2L}}{\mathrm{d}\log q} & \frac{\mathrm{d}\log p_I}{\mathrm{d}\log q} \end{pmatrix}', \quad b = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}'$$

In response to an exogenous shock $\mathrm{d}\log q$, we have

$$\frac{\partial \log K_{t+1}}{\partial \log q_t} = \delta \cdot \begin{pmatrix} \beta_K & \beta_{1L} & \beta_{2L} \end{pmatrix} A(\theta)^{-1} b$$

To obtain long-run dynamics, we set $\theta = 0$, which only affects the last row of $A(\theta)$ (i.e., $\mathrm{d}\log q = \mathrm{d}\log p_I$ in the long-run). In response to a permanent exogenous shock $\mathrm{d}\log q$,

$$\frac{\partial \log K}{\partial \log q} = \begin{pmatrix} \beta_K & \beta_{1L} & \beta_{2L} \end{pmatrix} A(0)^{-1} b$$

Capital demand elasticity. The demand elasticity is given by (steady-state + arbitrage)

$$\begin{aligned}
\mathrm{d}\log q &= (\alpha_K - 1)\,\mathrm{d}\log K + (\alpha_K - 1)\,\mathrm{d}\log(1-s_K) + \alpha_{1L}\,\mathrm{d}\log(1-s_{1L}) + \alpha_{2L}\,\mathrm{d}\log(1-s_{2L}) \\
&= (\alpha_K - 1)\,\mathrm{d}\log K - (\alpha_K - 1)\frac{s_K}{1-s_K}\,\mathrm{d}\log s_K - \alpha_{1L}\frac{s_{1L}}{1-s_{1L}}\,\mathrm{d}\log s_{1L} - \alpha_{2L}\frac{s_{2L}}{1-s_{2L}}\,\mathrm{d}\log s_{2L}
\end{aligned}$$

Therefore, we have that

$$\begin{aligned}\frac{\mathrm{d}\log K}{\mathrm{d}\log q} &= -\frac{1}{1-\alpha} + \frac{s_K}{1-s_K}\frac{\mathrm{d}\log s_K}{\mathrm{d}\log q} - \frac{\alpha_{1L}}{1-\alpha_K}\frac{s_{1L}}{1-s_{1L}}\frac{\mathrm{d}\log s_{1L}}{\mathrm{d}\log q} - \frac{\alpha_{2L}}{1-\alpha_K}\frac{s_{2L}}{1-s_{2L}}\frac{\mathrm{d}\log s_{2L}}{\mathrm{d}\log q} \\ &= -\frac{1}{1-\alpha} + \begin{pmatrix} \frac{s_K}{1-s_K} & -\frac{\alpha_{1L}}{1-\alpha_K}\frac{s_{1L}}{1-s_{1L}} & -\frac{\alpha_{2L}}{1-\alpha_K}\frac{s_{2L}}{1-s_{2L}} & 0 \end{pmatrix} A(0)^{-1} b\end{aligned}$$

C.2 Tax experiment

Long-run. The capital income tax rate $1-\tau_K$ represents a pure demand shock.

$$\mathrm{d}\log K = \epsilon^S \,\mathrm{d}\log q \tag{supply}$$

$$\mathrm{d}\log K = -\epsilon^D \,\mathrm{d}\log q - \mathrm{d}\log(1-\tau) \tag{demand}$$

The price response is

$$\frac{\mathrm{d}\log q}{\mathrm{d}\log(1-\tau)} = -\frac{1}{1-\alpha_K}\cdot\frac{1}{\epsilon^S+\epsilon^D}$$

The quantity response is

$$\frac{\mathrm{d}\log K}{\mathrm{d}\log(1-\tau)} = -\frac{1}{1-\alpha_K}\cdot\frac{\epsilon^S}{\epsilon^S+\epsilon^D}$$

Short-run partial equilibrium. Consider a partial equilibrium capital income tax shock $\mathrm{d}(1-\tau_t) = (1-\rho)^t\,\mathrm{d}(1-\tau)$ at $t=0$ around the undistorted steady-state. Using the envelope theorem on the firm value function, and using the fact that $\mathrm{d}w_{nL,t}=0$, we obtain

$$\begin{aligned}\mathrm{d}\log V_0 &= \frac{\frac{1}{1-\rho}\sum_{t=1}^{\infty}\left(\frac{1-\rho}{1+r}\right)^t D}{\sum_{t=1}^{\infty}\left(\frac{1}{1+r}\right)^t D}\cdot\mathrm{d}\log(1-\tau) \\ &= \frac{r}{r+\rho}\cdot\mathrm{d}\log(1-\tau)\end{aligned}$$

Using the fact that K_1 is pre-determined, we have

$$\mathrm{d}\log q_0 = \frac{r}{r+\rho}\,\mathrm{d}\log(1-\tau).$$

Using (investment-q), we have that

$$\mathrm{d}\log\frac{p_{I,0}I_0}{K_0} = \frac{1}{\theta}\,\mathrm{d}\log q_0$$

Using the fact that K_0 is predetermined, we have that

$$\mathrm{d}\log\left(p_{I,0}I_0\right) = \frac{1}{\theta}\frac{r}{r+\rho}\,\mathrm{d}\log(1-\tau)$$

We can alternatively derive an expression for $\mathrm{d}q_0$ using the fact that $\mathrm{d}r_K = 0$.

$$\begin{aligned} R\,\mathrm{d}q &= \mathrm{d}(1-\tau)r_K + \theta\,\mathrm{d}\left(\frac{I'}{K'}\right)^{1-\theta} q' + (1-(1-\theta)\delta)\,\mathrm{d}q', \\ &= \mathrm{d}(1-\tau)r_K + (1-\theta)\delta\,\mathrm{d}q' + (1-(1-\theta)\delta)\,\mathrm{d}q', \\ &= \mathrm{d}(1-\tau)r_K + \mathrm{d}q'. \end{aligned}$$

Therefore, we get that $\mathrm{d}\log q_0 = \frac{r}{r+\rho}\,\mathrm{d}\log(1-\tau)$.